SEQRET: Mining Rule Sets from Event Sequences

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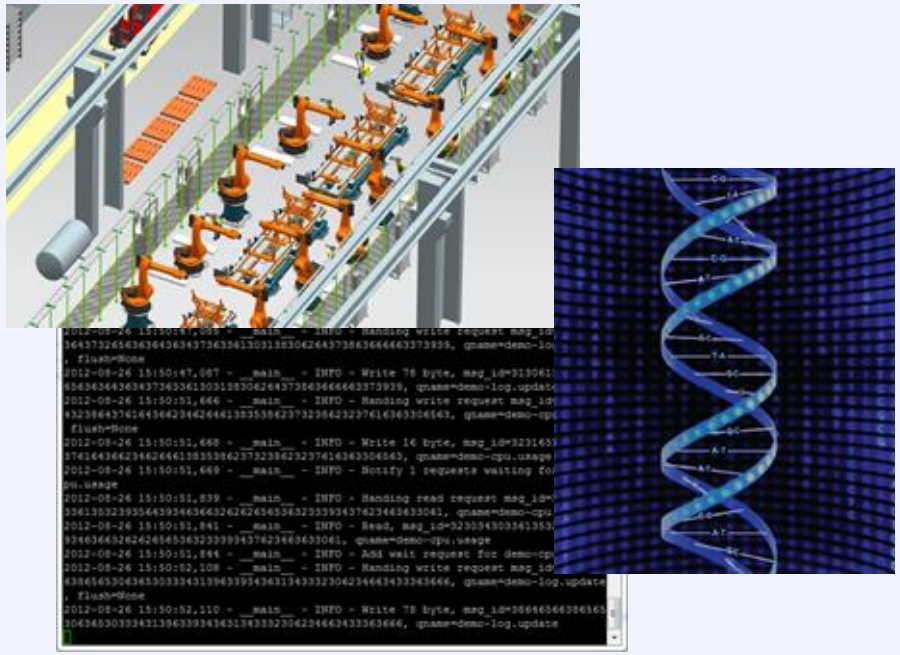
What are Event Sequences? What are Rules? Why are we interested in them?



Discrete sequences of ordered entities called events.
For example: <A, B, C, D, A, B, E, F, A, G>

Rules capture conditional dependencies between events.[1]

A ⟶ B indicates that B tends to occur following A.

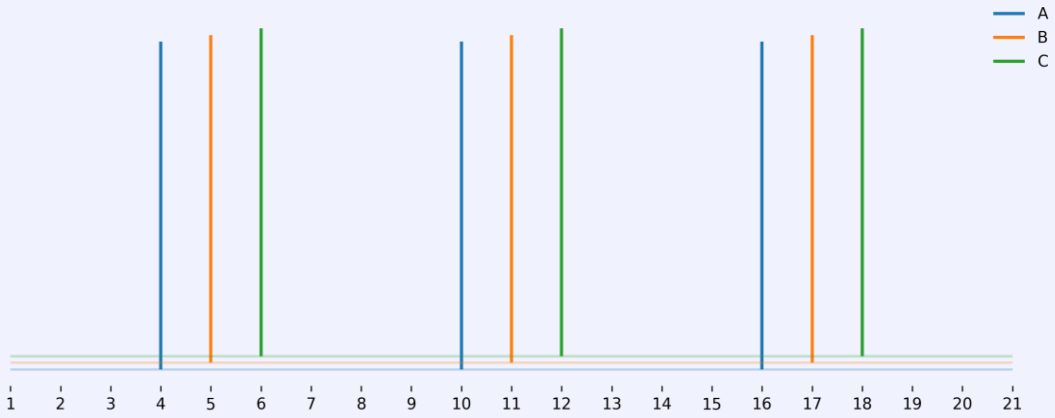
Rules are useful in predicting future events and in discovering underlying data generating mechanisms.

1. Fischer and Vreeken, 2020

How do we identify Rules?

Using information of order and context:

A B C

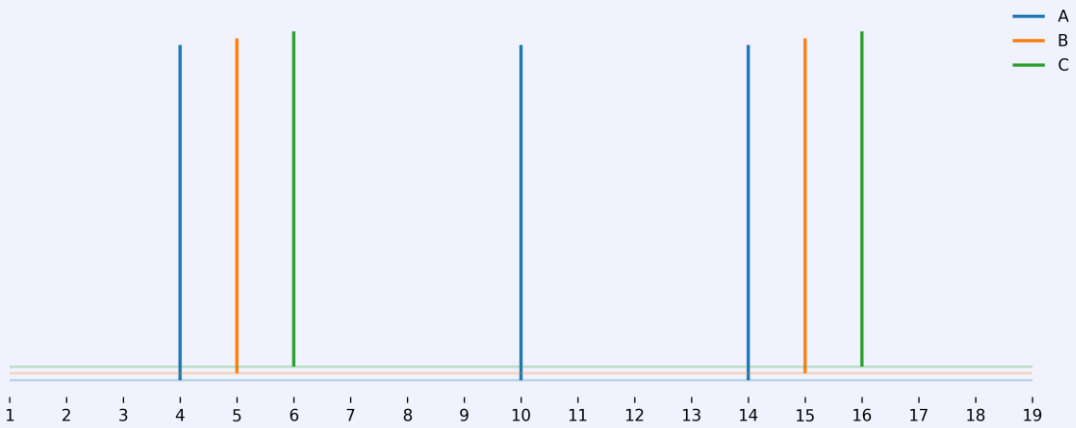


. . . A B C . . . A B C . . . A B C . . .

How do we identify Rules?

Using information of order and context:

A → B C



. . . A B C . . . A . . . A B C . . .

How do we identify Rules?

Using information of order and context:

A → B

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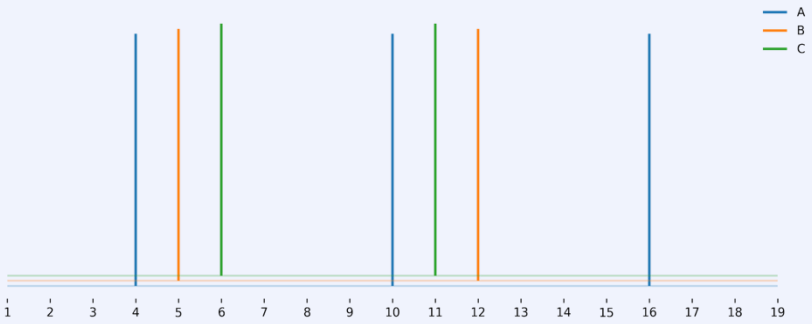
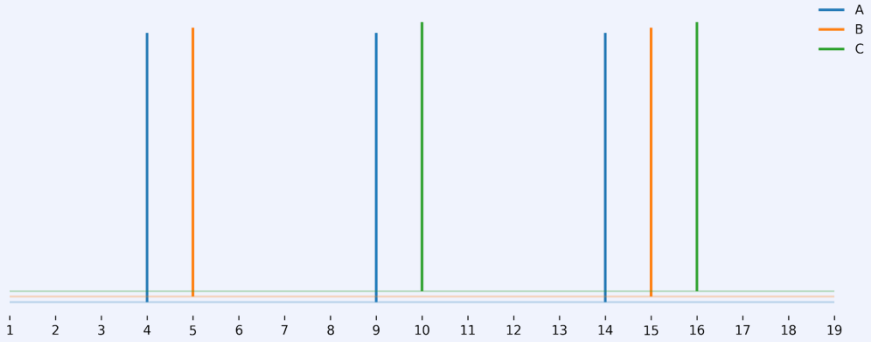


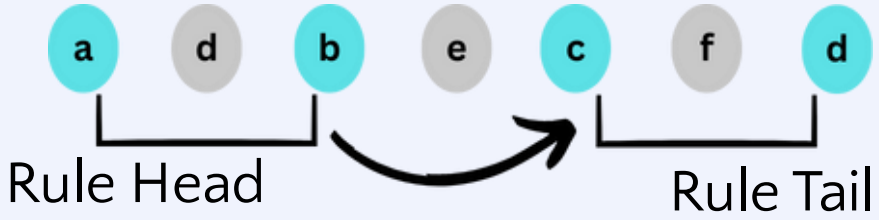
. . . A B C . . . A C B . . . A . . .



. . . A B . . . A C . . . A B C . . .

Rule Language

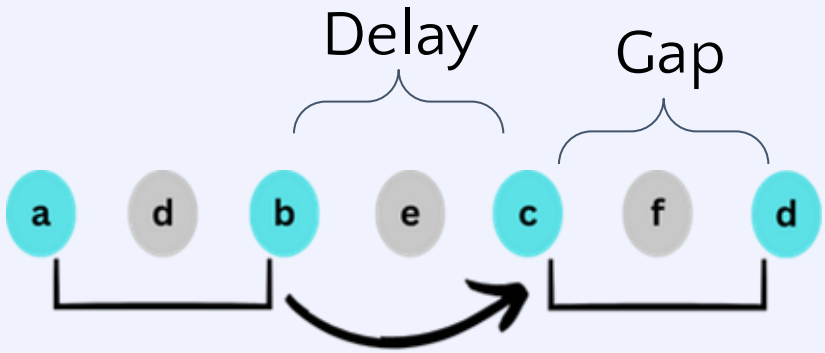
Fully-ordered rules between sequential patterns:



Independent patterns are represented as empty-head rules: $\varnothing \longrightarrow$ ABC

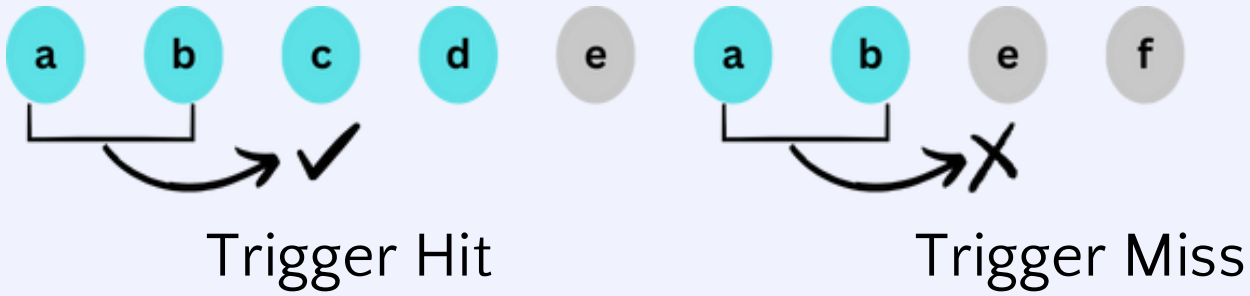
Rule Language

Gaps and delays allowed between the occurrences of constituent events:



Rule Language

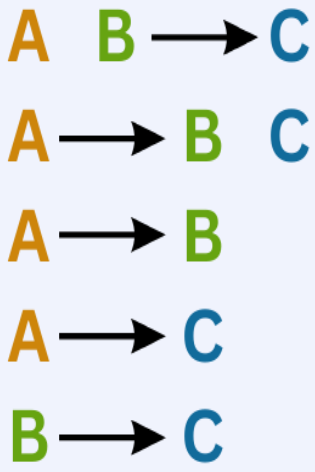
A rule triggers when its rule head is observed, at the position of the last event of the rule head:



Pattern Explosion Problem

- Existing solutions mine all rules that satisfy a minimum threshold for frequency, confidence or other metric of interest

- This results in exponentially many rules when the threshold is set too low and hardly any rules when the threshold is set too high



Minimum Description Length Principle to mine a **Rule Set**

Subset of possible rules

Description Length of the Data using the Model

how general are the rules?

$$\textbf{Find model M, minimizing } L(D, M) = L(M) + L(D \mid M)$$

from a class of models *M*.

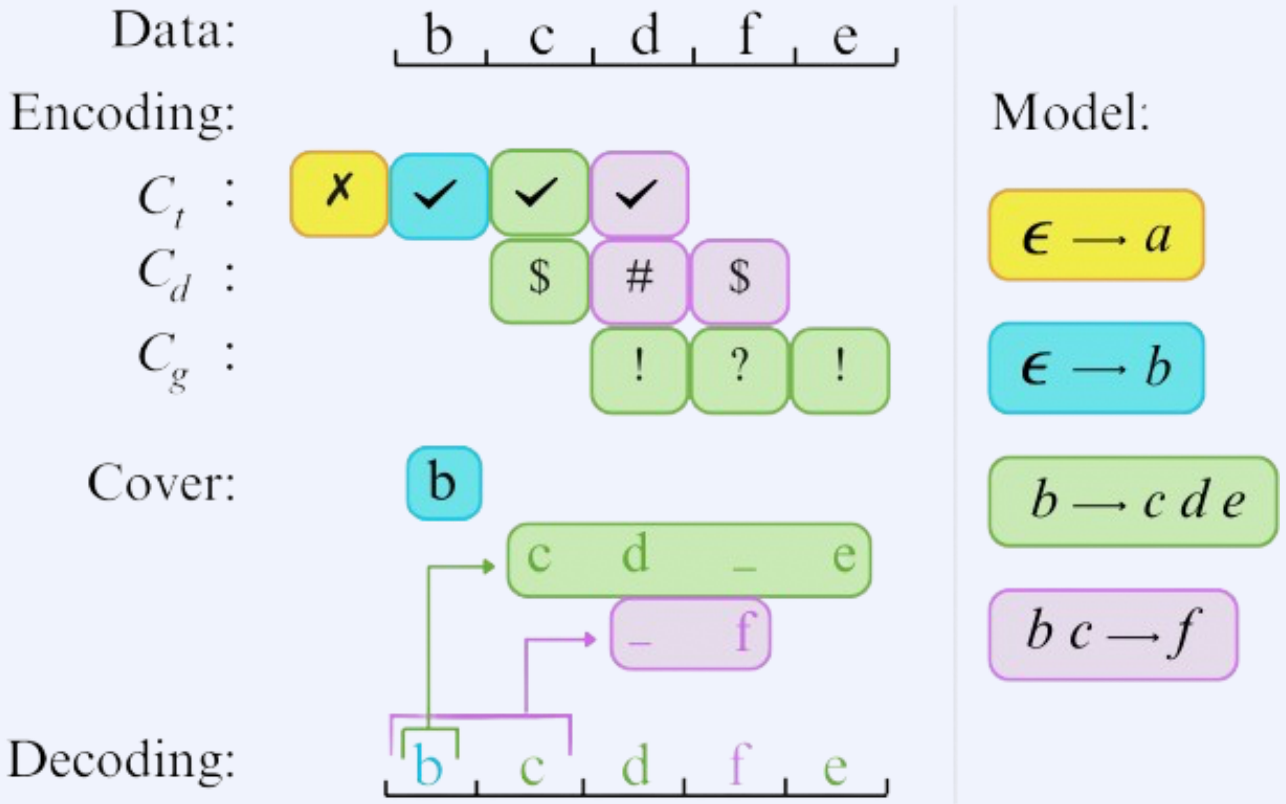
how complex are the rules?

Powerset of possible rules

Description Length of the Model

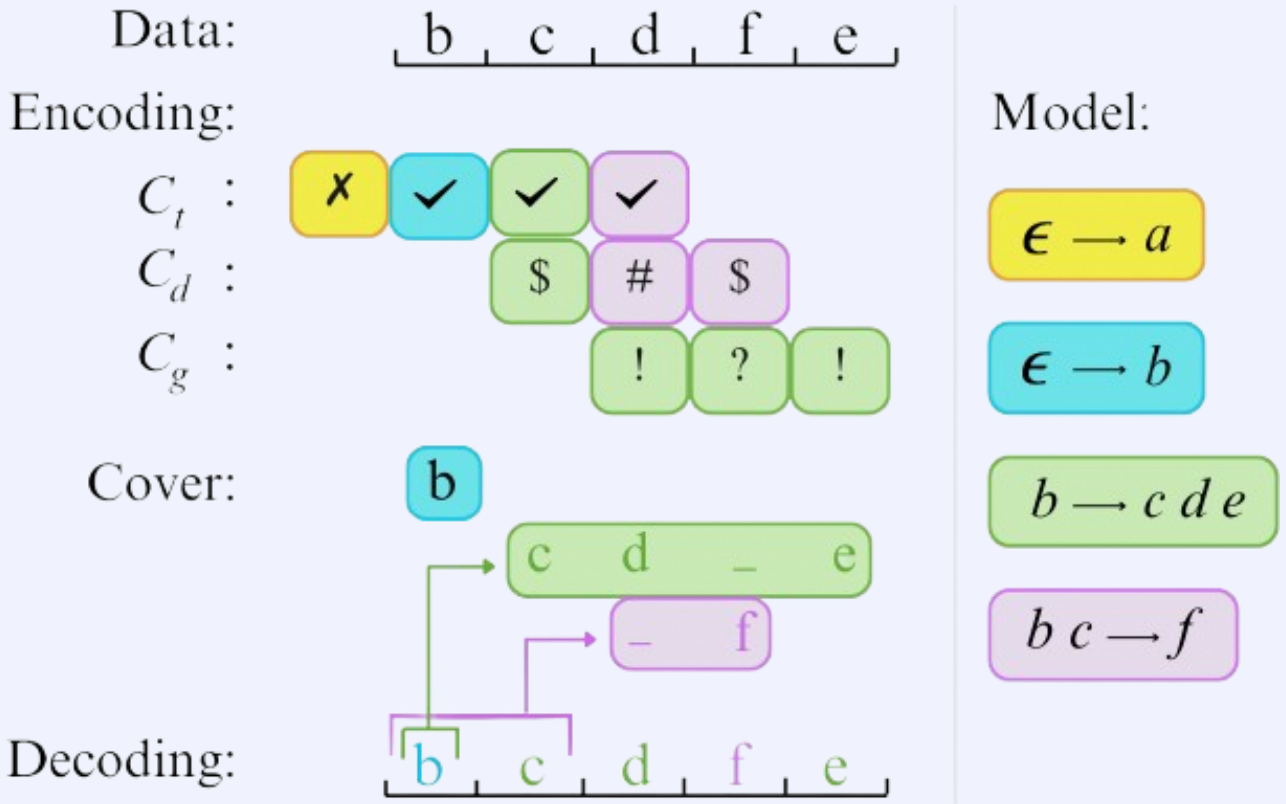
Encode the Data as Rule Tails

Given a rule set (model), a cover maps each event in the sequence to a rule tail.



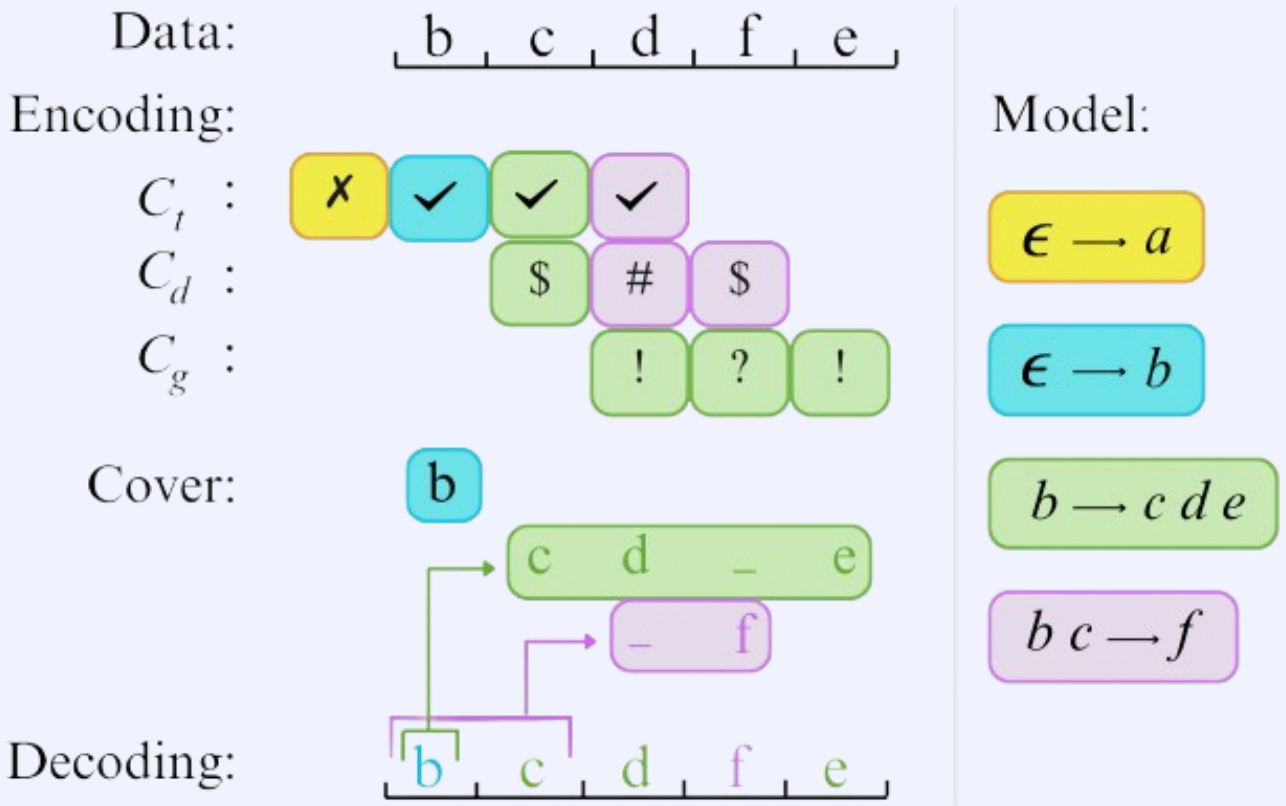
Encode the Data as Rule Tails

- Trigger stream C_t encodes whether a rule is triggered or not.
- Delay stream C_d encodes whether the rule tail immediately follows or not.
- Gap stream C_g encodes whether a constituent event of the rule tail occupies current position or not.



Encode the Data as Rule Tails

Finding the optimal cover (minimum encoding cost) is computationally expensive. Hence, we resort to a greedy cover, e.g. cover with the longest rule tail first.



SEQRET-MINE: Sequential Rule Set Miner

Encoding cost of the data as per the greedy cover = Description Length of the Data using the Model

how general are the rules?

Find model M, minimizing $L(D, M) = L(M) + L(D \mid M)$

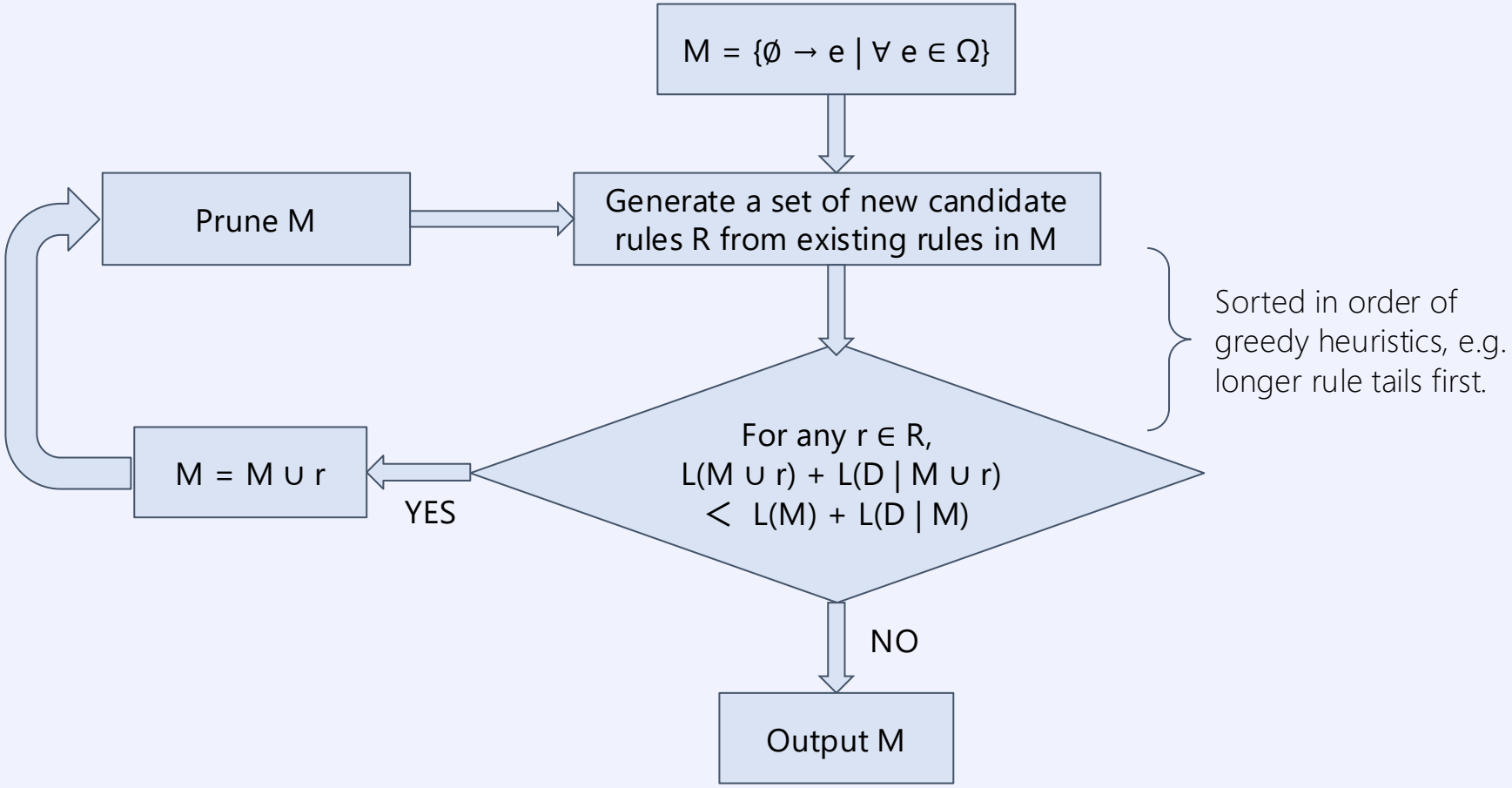
from a class of models *M*.

how complex are the rules?

Description Length of the Model

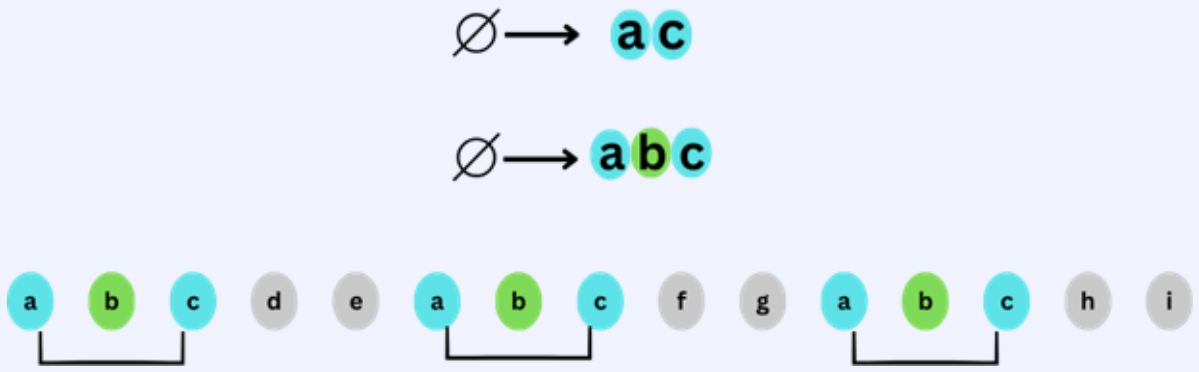
Mine that rule set which minimizes the total encoding cost!

SEQRET-MINE: **Seq**uential **R**ule S**et** Miner



SEQRET-MINE: Sequential Rule Set Miner

Candidate Rules Generation: Extend rules with events that occur in its neighborhood, more often than expected (Poisson Binomial Distribution).



Results on Synthetic Datasets

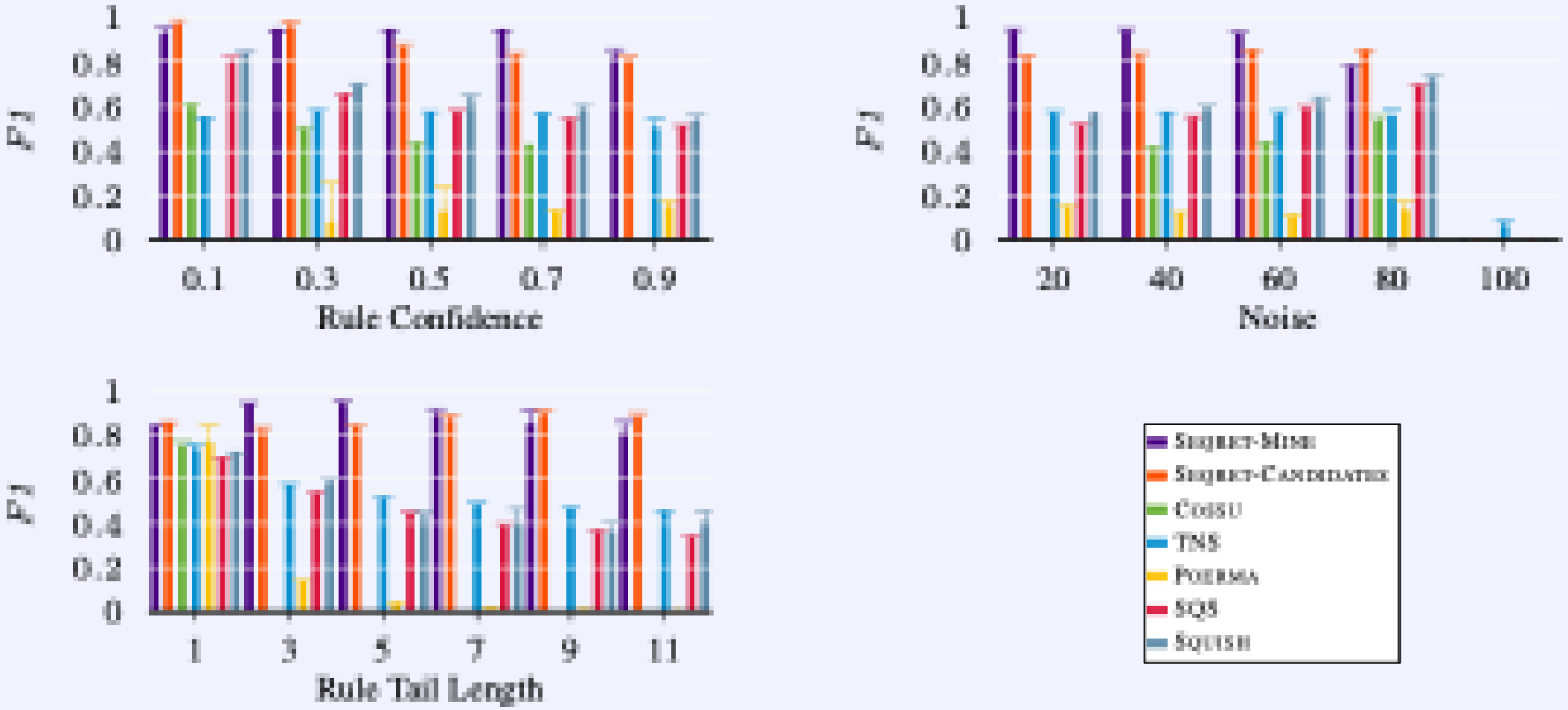
How similar is the mined rule set to the ground truth?

Measure similarity between rules as a weighted sum of the similarity between their component patterns:

$$s(A \rightarrow B, C \rightarrow D) = 0.5 \times s(AB, CD) + 0.25 \times s(A, C) + 0.25 \times s(B, D)$$

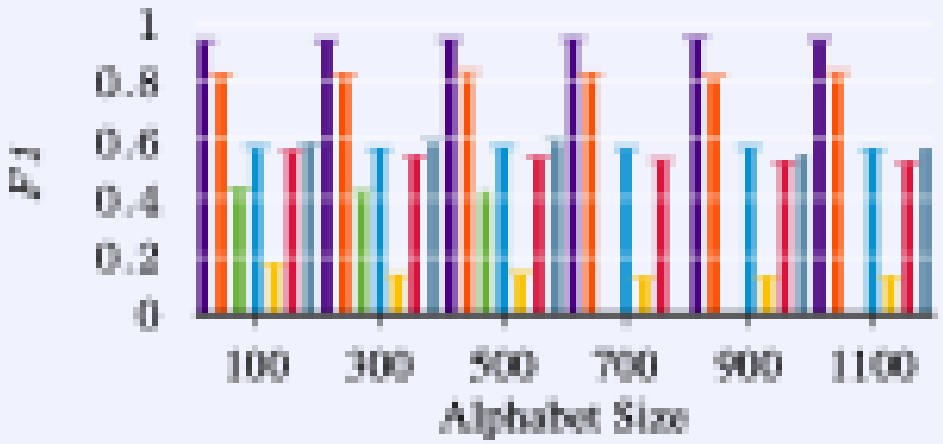
Similarity score based on LCS edit distance.

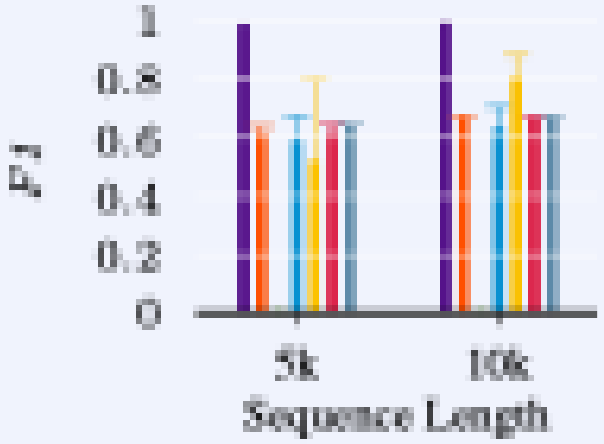
Results on Synthetic Datasets

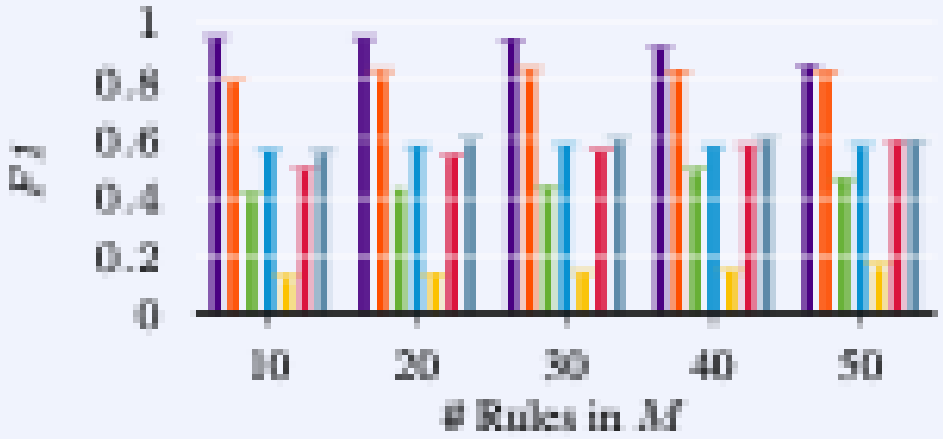


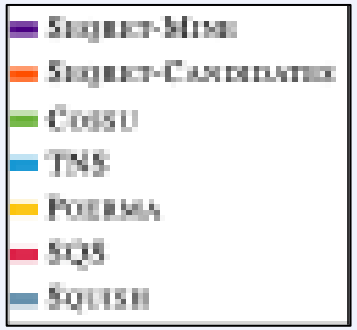
Results on Synthetic Datasets

Rules heads are not planted, only as frequent expected at random.



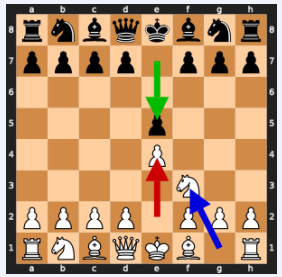






Results on Real-world Datasets

Lichess Games



$\varnothing \longrightarrow$ White:e4, Black:e5

White:e4, Black:e5 $\longrightarrow$ White:Nf3

Presidential Addresses

Public $\longrightarrow$ Debt

Public $\longrightarrow$ Money

Men $\longrightarrow$ Women, Children

Thank you!

We make available the code, data and an extended version of the paper at

https://eda.rg.cispa.io/prj/seqret/

https://github.com/alijosephine/SEQRET